Chapter 8. Inference in Multiple Regression II

Extra sums of squares, partial and sequential F, joint regions, simultaneous intervals, equality of regressions

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8.1 Extra sums of squares

When regressors are added to a model, SSR increases (and SSE decreases). The increase is the **extra sum of squares** due to the added regressors, and it measures their contribution *given* those already present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$

$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

Theorem 8.1

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of SSE.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

Proof. (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under H_0, $\lambda = 0$. ■

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

When the intercept is included in the regression sum of squares, one uses the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ with $p+1$ degrees of freedom and $SST = \mathbf{y}'\mathbf{y}$ with n; then $MSR/MSE \sim F(p+1, n-p-1)$ tests $\beta = \mathbf{0}$.

8.2 Partial and sequential F tests

Partial F test. The contribution of a single regressor x_j given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial F test**. It is equivalent to the t test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \Big(\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}\Big)^2 = t_0^2.$$

(This follows from Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial F means that x_j adds little *once the other variables are in the model*; it does not mean that x_j is unrelated to y. With correlated regressors, every partial F can be small while the overall F is large.

Sequential F tests. Enter the regressors one at a time in a chosen order, and at each stage test the newly added variable given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to SSR, and each is compared with the MSE of the model at that stage (or of the full model). The decomposition depends on the order unless the centred regressors are mutually orthogonal. The tests are valid only if the MSE used is unbiased for σ^2, which requires that the omitted variables have no effect; otherwise the denominator is inflated.

8.3 Joint confidence regions and simultaneous intervals

Motivation. Several individual 95% intervals do not hold simultaneously with probability 95%. Joint regions and Bonferroni or Scheffé intervals control the error rate for all statements at once.

Theorem 8.2 (Joint confidence region)

Under normal errors,

$$\frac{(\hat\beta-\beta)'X'X(\hat\beta-\beta)}{(p+1)MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta-\beta)'X'X(\hat\beta-\beta) \le (p+1)MSE\, F_\alpha(p+1,n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for β.

Proof. $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta-\beta)'X'X(\hat\beta-\beta) = \epsilon' H \epsilon$, and $\epsilon' H\epsilon/\sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I-H)\epsilon$, since $H(I-H)=0$. ∎

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

Theorem 8.3 (Bonferroni and Scheffé intervals)

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj}MSE}$, $j = 0, \ldots, p$, cover all β_j simultaneously with probability at least $1-\alpha$.

(b) (Scheffé) With probability $1-\alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1)F_\alpha(p+1,n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

Proof. (a) If A_j is the event that the j-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1-\alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a}\neq\mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ■

Taking $\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole regression surface, valid for all $\mathbf{x}$ at once. For a few coefficients Bonferroni intervals are shorter; Scheffé intervals pay for covering infinitely many combinations.

8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with n_1 and n_2 observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* σ^2. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n-2(p+1)$ degrees of freedom $(n = n_1 + n_2)$.

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one β_0: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under H_0,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (with block-diagonal design $\mathrm{diag}(X_1, X_2)$ and parameter $(\beta_1', \beta_2')'$) under the $p+1$ linear restrictions $\beta_1 - \beta_2 = \mathbf{0}$, so this is again a general linear hypothesis. For k groups and $H_0 : \beta_1 = \cdots = \beta_k$, the numerator has $(k-1)(p+1)$ degrees of freedom and the denominator $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.

8.5 Selected exercises

Motivation. The exercises derive the equality test of Section 8.4 from the difference of the two fitted coefficient vectors, rewrite partial F statistics through R^2, show when sequential sums of squares do not depend on the order, and compare Scheffé and t intervals.

Exercise 8.1 (Equality of two regressions via $\hat\beta_1 - \hat\beta_2$)

In the setting of Section 8.4, let $\hat\beta_1, \hat\beta_2$ be the separate least squares estimators, $G_i = (X_i'X_i)^{-1}$, $\hat\Delta = \hat\beta_1 - \hat\beta_2$ and $MSE = (SSE_1 + SSE_2)/(n-2p-2)$.
(a) Find the distribution of $\hat\Delta$, and show that $(n-2p-2)MSE/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.
(b) Show that under $H_0 : \beta_1 = \beta_2$,

$$F_1 = \frac{\hat\Delta'(G_1 + G_2)^{-1}\hat\Delta/(p+1)}{MSE} \sim F(p+1, n-2p-2).$$

(c) Show that F_1 equals the statistic F_0 of Section 8.4.

Solution.

(a) By Theorem 6.3, $\hat\beta_i \sim N_{p+1}(\beta_i, \sigma^2 G_i)$. The two samples are independent, so $\hat\beta_1$ and $\hat\beta_2$ are independent and $\hat\Delta \sim N_{p+1}(\beta_1-\beta_2, \sigma^2(G_1+G_2))$. By Theorem 6.4, $SSE_i/\sigma^2 \sim \chi^2(n_i-p-1)$ independently of $\hat\beta_i$; together with the independence of the samples, the four quantities $\hat\beta_1, SSE_1, \hat\beta_2, SSE_2$ are mutually independent. Hence $(SSE_1+SSE_2)/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.

(b) $\Sigma = \sigma^2(G_1+G_2)$ is positive definite. Under H_0, $\hat\Delta \sim N(\mathbf{0}, \Sigma)$, and with $A = \Sigma^{-1}$ we have $A\Sigma = I$, which is idempotent of rank $p+1$; Theorem 5.5(a) gives $\hat\Delta'\Sigma^{-1}\hat\Delta \sim \chi^2(p+1)$. Dividing by $(n-2p-2)MSE/\sigma^2$ and using (a),

$$F_1 = \frac{\hat\Delta'\Sigma^{-1}\hat\Delta/(p+1)}{\{(n-2p-2)MSE/\sigma^2\}/(n-2p-2)} \sim F(p+1, n-2p-2).$$

(c) Write the full model as one regression with design $X_F = \operatorname{diag}(X_1, X_2)$ and parameter $\mathbf{b} = (\beta_1', \beta_2')'$. Then $X_F'X_F = \operatorname{diag}(X_1'X_1, X_2'X_2)$, so the normal equations decouple, $\hat{\mathbf{b}} = (\hat\beta_1', \hat\beta_2')'$ and $SSE(F) = SSE_1 + SSE_2$. The hypothesis is $C\mathbf{b} = \mathbf{0}$ with $C = (I_{p+1}, -I_{p+1})$, and

$$C(X_F'X_F)^{-1}C' = G_1 + G_2, \qquad C\hat{\mathbf{b}} = \hat\Delta, \qquad Q = \hat\Delta'(G_1+G_2)^{-1}\hat\Delta.$$

By Theorem 7.3 (Method II), $SSE(R) - SSE(F) = Q$, so $F_0 = \{Q/(p+1)\}/MSE = F_1$. ■

Exercise 8.2 (Extra sums of squares through R^2)

Let the full model (with intercept) have p regressors and coefficient of determination R_F^2, and the submodel the first q of them, with R_R^2.
(a) Show that the F statistic of Theorem 8.1(c) equals

$$F_0 = \frac{(R_F^2 - R_R^2)/(p-q)}{(1-R_F^2)/(n-p-1)}.$$

(b) Show that $R_F^2 \geq R_R^2$, with equality if and only if $\hat\beta_{q+1} = \cdots = \hat\beta_p = 0$ in the full fit.

Solution [Supplement].

(a) Both models contain the intercept and are fitted to the same $\mathbf{y}$, so they share SST, and $SSE = SST(1-R^2)$ for each. Hence $SS(\beta_2 \mid \beta_1) = SSE(\text{sub}) - SSE(\text{full}) = SST(R_F^2 - R_R^2)$ and $MSE = SST(1-R_F^2)/(n-p-1)$; SST cancels in the ratio.

(b) $R_F^2 - R_R^2 = \mathbf{y}'(H-H_1)\mathbf{y}/SST = \|(H-H_1)\mathbf{y}\|^2/SST \geq 0$, because $H - H_1$ is symmetric idempotent (Theorem 8.1(a)). By Theorem 6.5, the full-model residual is $(I-H)\mathbf{y} = (I-H_1)\mathbf{y} - X_{2\cdot1}\hat\beta_2$, so

$$(H-H_1)\mathbf{y} = (I-H_1)\mathbf{y} - (I-H)\mathbf{y} = X_{2\cdot1}\hat\beta_2.$$

$X_{2\cdot1} = (I-H_1)X_2$ has full column rank: if $X_{2\cdot1}\mathbf{a} = \mathbf{0}$, then $X_2\mathbf{a} = H_1X_2\mathbf{a}$ lies in the column space of X_1, contradicting the full rank of X unless $\mathbf{a} = \mathbf{0}$. Therefore $R_F^2 = R_R^2$ iff $X_{2\cdot1}\hat\beta_2 = \mathbf{0}$ iff $\hat\beta_2 = \mathbf{0}$. ■

Exercise 8.3 (When the order of entry does not matter)

Partition $X = (X_1, X_2)$ as in Section 8.1, X_1 containing the intercept, and suppose $X_1'X_2 = 0$ (for instance, the columns of X_2 are centred and uncorrelated with those of X_1). Let $H_2 = X_2(X_2'X_2)^{-1}X_2'$.
(a) Show that $H = H_1 + H_2$, and hence $SS(\beta_2 \mid \beta_1) = \mathbf{y}'H_2\mathbf{y}$, which does not involve X_1.
(b) Show that $\hat\beta_2 = (X_2'X_2)^{-1}X_2'\mathbf{y}$.
(c) Deduce that if the centred regressors are split into mutually orthogonal blocks, the sequential sums of squares of the blocks do not depend on the order of entry.

Solution [Supplement].

(a) $H_1H_2 = X_1(X_1'X_1)^{-1}(X_1'X_2)(X_2'X_2)^{-1}X_2' = 0$, and likewise $H_2H_1 = 0$. So $P = H_1 + H_2$ is symmetric with $P^2 = H_1 + H_2 = P$, an orthogonal projection. Its range contains the columns of X_1 and X_2 ($PX_1 = X_1$, $PX_2 = X_2$) and is contained in the column space of X, so it equals the column space of X, and $P = H$ by uniqueness of the orthogonal projection. Then $SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y} = \mathbf{y}'H_2\mathbf{y}$.

(b) $(I - H_1)X_2 = X_2 - X_1(X_1'X_1)^{-1}X_1'X_2 = X_2$, so $X_{2\cdot 1} = X_2$, and by Theorem 6.5, $\hat\beta_2 = (X_2'X_2)^{-1}X_2'(I - H_1)\mathbf{y} = (X_2'X_2)^{-1}X_2'\mathbf{y}$, using $X_2'H_1 = 0$.

(c) Let the centred blocks $Z_1, \dots, Z_m$ satisfy $Z_j'Z_k = 0$ for $j \neq k$ and $\mathbf{1}'Z_j = \mathbf{0}$. Whatever blocks have entered before Z_j, together with the intercept they form an X_1 orthogonal to Z_j, so by (a) the sequential sum of squares of Z_j is $\mathbf{y}'H_{Z_j}\mathbf{y}$ in every order. These sum to SSR, by (a) applied repeatedly. ∎

Exercise 8.4 (Scheffé intervals are wider than t intervals)

Let $\nu = n - p - 1$. Show that $(p+1)F_\alpha(p+1, \nu) \geq t_{\alpha/2}(\nu)^2$, so that for a single combination $\mathbf{a}'\beta$ the Scheffé interval of Theorem 8.3(b) contains the t interval of Theorem 7.2(b).

Solution [Supplement].

Step 1. Fix $\mathbf{a} \neq \mathbf{0}$ and let $T_{\mathbf{a}} = \mathbf{a}'(\hat\beta - \beta)/\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}$. By Theorem 7.2, $T_{\mathbf{a}} \sim t(\nu)$, so $T_{\mathbf{a}}^2 \sim F(1, \nu)$, whose distribution function is continuous and strictly increasing on $(0, \infty)$.

Step 2. By the proof of Theorem 8.3(b), $\sup_{\mathbf{b} \neq \mathbf{0}} T_{\mathbf{b}}^2 = (\hat\beta - \beta)'X'X(\hat\beta - \beta)/MSE$, and by Theorem 8.2 this is at most $c = (p+1)F_\alpha(p+1, \nu)$ with probability $1 - \alpha$.

Step 3. Since $T_{\mathbf{a}}^2 \leq \sup_{\mathbf{b}} T_{\mathbf{b}}^2$, $P(T_{\mathbf{a}}^2 \leq c) \geq 1 - \alpha = P(T_{\mathbf{a}}^2 \leq t_{\alpha/2}(\nu)^2)$. By the strict monotonicity in Step 1, $c \geq t_{\alpha/2}(\nu)^2$. The half-widths are $\sqrt{c}$ and $t_{\alpha/2}(\nu)$ times the same standard error, which proves the claim. ∎